Reducing False Positives in CAN Fuzzing through Reproducibility-Aware Validation and Multi-Monitor Integration*

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Abstract

Real-vehicle CAN fuzzing environments exhibit non-deterministic ECU behavior and black-box characteristics, making single-execution analysis unreliable.

This study proposes a multi-monitor CAN fuzzing framework integrating Timing, DBC, and UDS information with reproduction-based verification to filter transient anomalies and identify actual failures.

Experimental results from 27 hours of fuzzing on an Audi A5 showed that, among 11 fail candidates, 5 were consistently reproduced, 2 were classified as partial failures, and 4 were classified as non-reproducible transient anomalies. To the best of our knowledge, this is among the first studies to combine multi-monitor anomaly detection with reproduction-based verification in real-vehicle CAN fuzzing.

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